

App Dev -1 Project report

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About- I am a student in the IIT Madras BS Degree Programme. I enjoy learning web application development and database management. This project helped me understand Flask, SQLAlchemy, role-based authentication, CRUD operations, and responsive web application development using Bootstrap.

Project Details

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Trekking Management Application

Problem Statement

The objective of this project is to develop a web-based Trekking Management Application that enables administrators, staff members, and customers to efficiently manage trekking activities.

Traditional trekking management often involves manual record keeping, making it difficult to manage trek schedules, staff assignments, bookings, and participant information. This application automates these processes by providing role-based access to different users while ensuring secure authentication and efficient data management.

Approach

The application was developed using the Flask framework following a modular architecture.

The project provides three different user roles:

- **Admin** manages treks, staff members, users, and bookings.
- **Staff** manages assigned treks by updating trek status, available slots, and maintaining their profile.
- **Customer** can browse available treks, search and filter treks, make bookings, cancel bookings, and view booking history.

The database was implemented using SQLite with SQLAlchemy ORM. Bootstrap was used to create a responsive and user-friendly interface.

AI / LLM Declaration

I used ChatGPT (GPT-5.5) during the development of this project for:

- Debugging Flask routes
- Improving Bootstrap-based user interface
- Optimizing HTML templates
- Understanding SQLAlchemy relationships
- Improving code structure
- Preparing the project report

The AI assistance was limited to guidance and code improvement. The application logic, integration, testing, debugging, and implementation were completed by me.

Technologies and Frameworks Used

Technology	Purpose
Flask	Backend web framework
SQLAlchemy	ORM for SQLite database
SQLite	Local database
Jinja2	Dynamic HTML templates
Bootstrap 5	Responsive user interface
HTML5	Frontend structure
CSS3	Styling
Werkzeug Security	Password hashing and authentication

Database Schema / ER Diagram

Tables

User

Stores user information.

Fields-id,name,username,password,role,active and approved

Trek

Stores trekking details.

Fields-id,name,location,difficulty,duration,price,available_slots,
assigned_staff_id,status

Booking

Stores booking information.

Fields-id,user_id,trek_id,booking_date,persons,total_price,
booking_status and payment_status

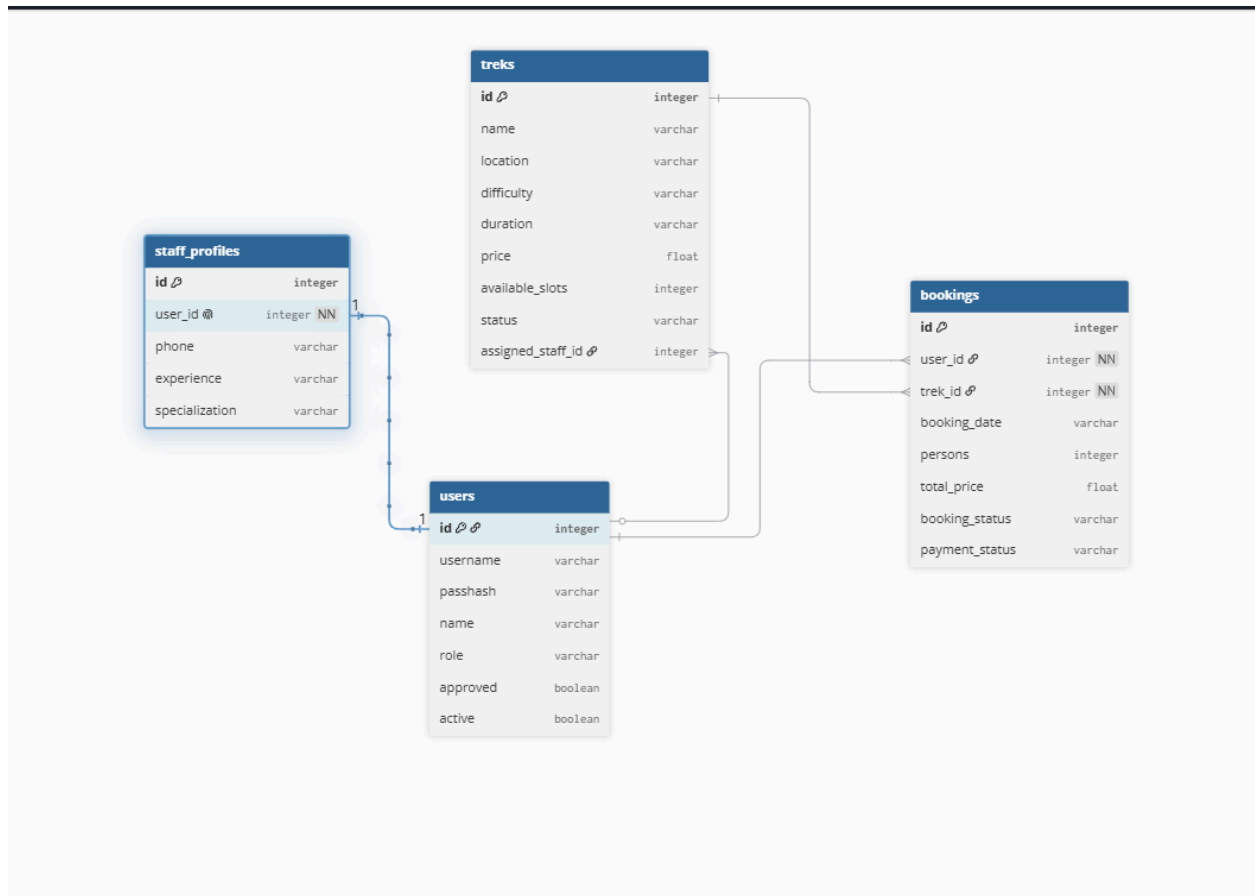
StaffProfile

Stores additional information related to staff members.

Fields-id,user_id,phone,experience,specialization

Relationships

- One User → Many Bookings
- One Trek → Many Bookings
- One Staff Member → Many Treks
- One Staff User → One Staff Profile



Architecture and Features

```
app.py
models.py
routes.py
templates/
  admin/
  staff/
  user/
static/
instance/
```


Video Presentation

Video Presentation

Google Drive Link-

<https://drive.google.com/file/d/1R4YiVWRRWVUibw6M2viTJH5O5Tp0Uvzx/view?usp=sharing>

GitHub Link-

https://github.com/23f3000176/Trekking_Management_App.git